Sparse Recovery and Compressed Sensing

A Model-Based Deep Learning Approach

Core question: How far can model-based structure take us — and when do we need data-driven flexibility?

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Based on: Chen, Liu, Wang, Yin — "Hyperparameter Tuning is All You Need for LISTA" · NeurIPS 2021

Seminar Roadmap

1 — Background
Motivation, LASSO, ISTA / FISTA

2 — Unrolling
LISTA, ALISTA, HyperLISTA

3 — Our Study
Scalar variants + training methods

4 — Results
Sparse recovery & design space

5 — Images
MNIST & FashionMNIST CS stress tests

Goal: systematically compare how much learning is needed when moving from ideal sparse signals to real compressed-sensing images.

Synthetic signals
m=250, n=500, s=50
Gaussian A, noiseless

Trained models
LISTA, ALISTA, HyperLISTA
+ 3 scalar designs (ours)

Image CS
MNIST & FashionMNIST pixel domain
ratios $\in \{0.25, 0.5\}$

Motivation & Problem Formulation

Sparse recovery appears everywhere:

Application	Measurements	Goal
MRI acceleration	k-space $(m \ll n)$	Image pixels
Radar	Compressed snapshots	Reflectivity
Hyperspectral	Spectral bands	Spatial cube
Compressed sensing	Random projections	Original signal

Why tractable? Natural signals have only $s \ll n$ non-zero components.

LASSO — the optimization backbone:

$$\hat{x} = \arg\min_x \tfrac{1}{2}\|b - Ax\|_2^2 + \lambda\|x\|_1$$

LASSO = Least Absolute Shrinkage and Selection Operator (Tibshirani, 1996)

Measurement model

$$b = Ax^* + \varepsilon$$

A ∈ ℝ^{m×n}, m ≪ n

x*: s-sparse signal

ε: additive Gaussian noise

Metrics: NMSE (dB) for synthetic signals;
NMSE + PSNR + SSIM for image CS.

Classical Baselines: ISTA and FISTA

ISTA — proximal gradient descent on the LASSO objective:

$$x^{(k+1)} = \mathcal{S}_\theta\left(x^{(k)} + \tfrac{1}{L}A^T\left(b - Ax^{(k)}\right)\right), \qquad L = \|A\|_2^2, \quad \theta = \lambda/L$$

FISTA — ISTA with Nesterov momentum, $O(1/k^2)$ convergence vs $O(1/k)$:

$$y^{(k)} = x^{(k)} + \tfrac{t_k - 1}{t_{k+1}}\left(x^{(k)} - x^{(k-1)}\right), \qquad x^{(k+1)} = \mathcal{S}_\theta\left(y^{(k)} + \tfrac{1}{L}A^T\left(b - Ay^{(k)}\right)\right)$$

Method	NMSE @ K=16	Error energy [%]	Params
ISTA	−5.3 dB	29.5%	0
FISTA	−11.0 dB	7.9%	0

At K=16 iterations both methods are far from converged.
ISTA needs 200+ iterations for −20 dB.

Can we make 16 steps count?

ISTA = Iterative Shrinkage-Thresholding Algorithm; FISTA = Fast ISTA (Beck & Teboulle, 2009)

Deep Unrolling: From Iterations to Layers

Key idea (Gregor & LeCun, 2010): unfold K ISTA iterations into a K-layer network and learn the update parameters from data.

Classical ISTA — parameters fixed

$$z^{(k)} = x^{(k)} + \tfrac{1}{L} A^T \left(b - A x^{(k)}\right)$$

$$x^{(k+1)} = \mathcal{S}_\theta\left(z^{(k)}\right)$$

Parameters $1/L$ and $\theta = \lambda/L$ are fixed constants.

Fixed — no learning

Unrolled LISTA — parameters learned

$$z^{(k)} = W_x^{(k)} x^{(k)} + W_y^{(k)} b$$

$$x^{(k+1)} = \mathcal{S}_{\theta_k}\left(z^{(k)}\right)$$

$W_x^{(k)}$, $W_y^{(k)}$, θ_k are learned per layer via BPTT.

~6 M parameters — learned end-to-end

MBDL principle: preserve the known algorithmic structure (gradient step + soft-threshold);
learn only what improves finite-depth reconstruction. Training minimizes $\|x(K) - x^*\|^2$ via BPTT.

BPTT = Backpropagation Through Time — end-to-end backpropagation through all K unrolled layers, as when training a recurrent network

The MBDL Ladder: LISTA → ALISTA → HyperLISTA

LISTA — learn all matrices per layer, ~6 M params:

$$x^{(k+1)} = \mathcal{S}_{\theta_k}\big(W_x^{(k)} x^{(k)} + W_y^{(k)} b\big)$$

ALISTA — W analytic, learn step sizes and thresholds, 32 params:

$$W^* = (AA^T)^{-1}A \quad \text{(column-normalized)}$$

$$x^{(k+1)} = \mathcal{S}_{\theta_k}\big(x^{(k)} + \gamma_k W^{*T}(b - Ax^{(k)})\big)$$

HyperLISTA — adaptive θ/β/p from residuals, 3 params (grid search, zero backprop):

$$\theta^{(k)} = c_1 \mu \|A^+(Ax^{(k)} - b)\|_1, \qquad \beta^{(k)} = c_2 \mu \|x^{(k)}\|_0$$

$$p^{(k)} = c_3 \log \frac{\|A^+ b\|_1}{\|A^+(Ax^{(k)} - b)\|_1}$$

Model	Params	NMSE	Error [%]
ISTA	0	−5.3 dB	29.5%
FISTA	0	−11.0 dB	7.9%
LISTA	6 M	−20.3 dB	0.93%
ALISTA	32	−29.8 dB	0.10%
HyperLISTA	**3**	**−61.4 dB**	**0.00007%**

NMSE on synthetic sparse recovery (m=250, n=500, s=50, noiseless, K=16 layers)

More structure → fewer params → better performance

This is the MBDL principle in action.

Our Design Space Study: What Should We Learn?

Research question: LISTA learns 6 M parameters; HyperLISTA uses 3 scalars. What is the minimal useful set of learnable components?

The ISTA gradient-correction update has two free scalar sequences:

$$x^{(k+1)} = \mathcal{S}_{\theta_k}\big(x^{(k)} + \gamma_k A^T(b - Ax^{(k)})\big)$$

We designed three models, each learning a different subset:

ThresholdISTA

16 params

Learns $\theta_1, \dots, \theta_K$ only

Step fixed at $\gamma = 1/L$

Init: $\theta_0 = \lambda/L$

StepISTA

16 params

Learns $\gamma_1, \dots, \gamma_K$ only

Threshold coupled: $\theta_k = \lambda\gamma_k$

Init: $\gamma_0 = 1/L$

StepThresholdISTA

32 params

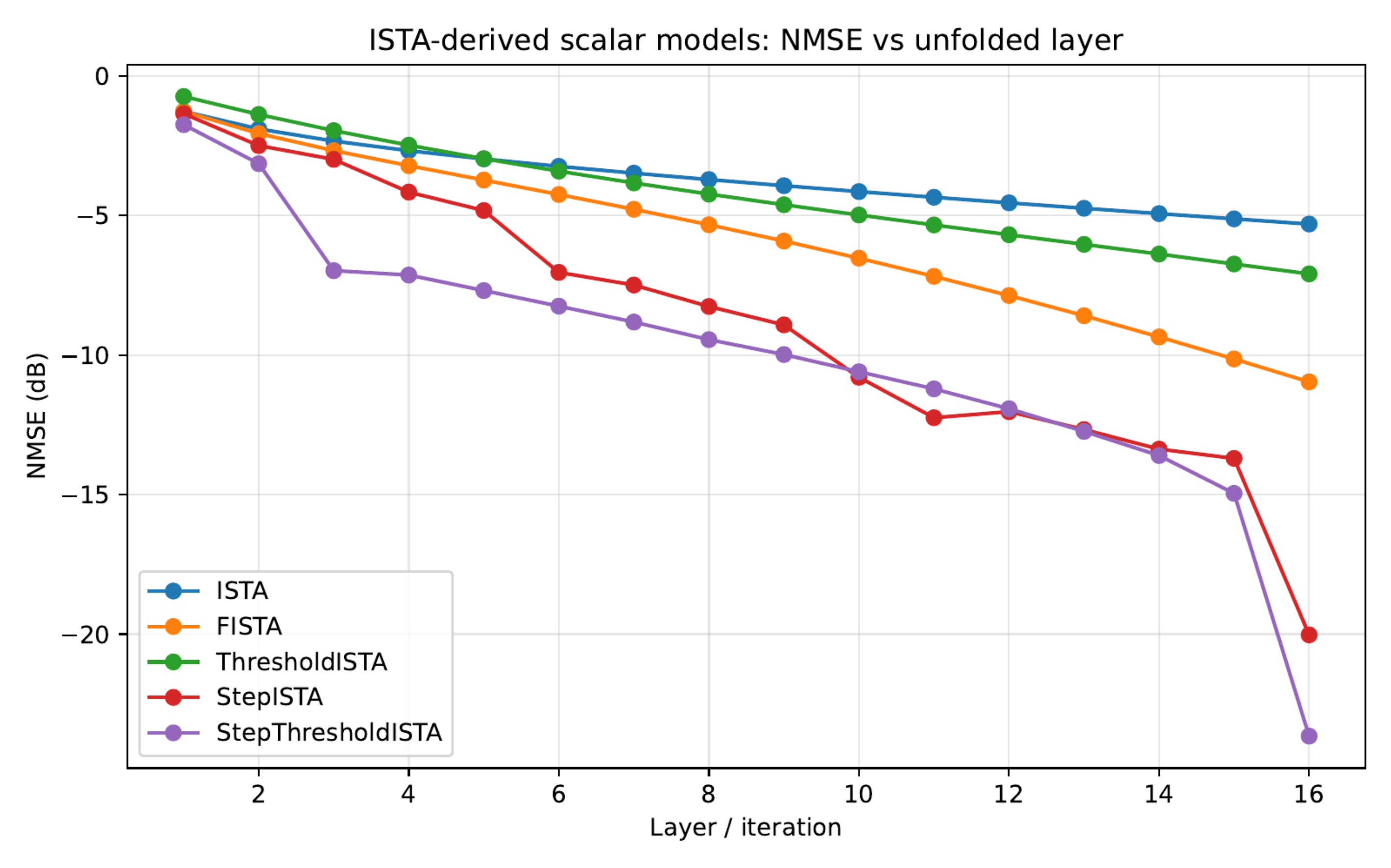
Learns both γ_k and θ_k

independently per layer

Init: ISTA values

All parameters softplus-reparameterized (positivity). All initialized to reproduce ISTA exactly before training.

Our Results: Scalar Models



Model	NMSE	Params
ISTA	−5.3 dB	0
FISTA	−11.0 dB	0
ThresholdISTA	−7.1 dB	16
StepISTA	**−20.0 dB**	**16**
StepThresholdISTA	−23.6 dB	32
LISTA-L1 (ref.)	−20.3 dB	6 M

Key finding

Learning γ_k alone (16 scalars) already matches full LISTA (6 M parameters).

Decoupling θ_k adds 3 dB at zero extra cost.

StepThresholdISTA beats LISTA-L1 with 187,000× fewer parameters.

Training Methods for Unfolded Optimizers

Architecture fixed (LISTA). Only the training strategy changes.

L1 — End-to-end

$$\mathcal{L} = \|x^{(K)} - x^*\|^2$$

Supervise final output only.
Standard BPTT.

L2 — Deep Supervision

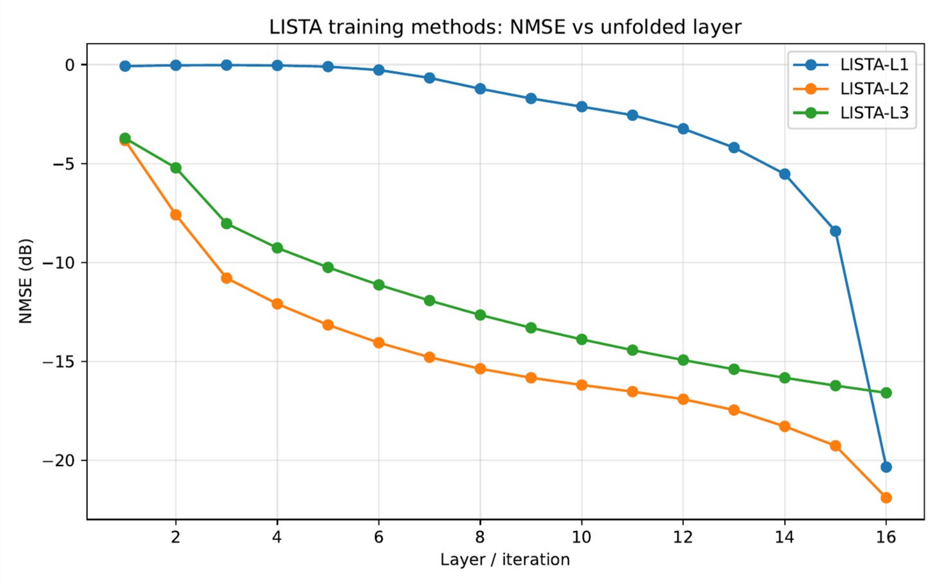
$$\mathcal{L} = \sum_k w_k \|x^{(k)} - x^*\|^2$$

Supervise every intermediate layer.

L3 — Greedy Layerwise

$$\min_{\theta_k} \|x^{(k)} - x^*\|^2$$

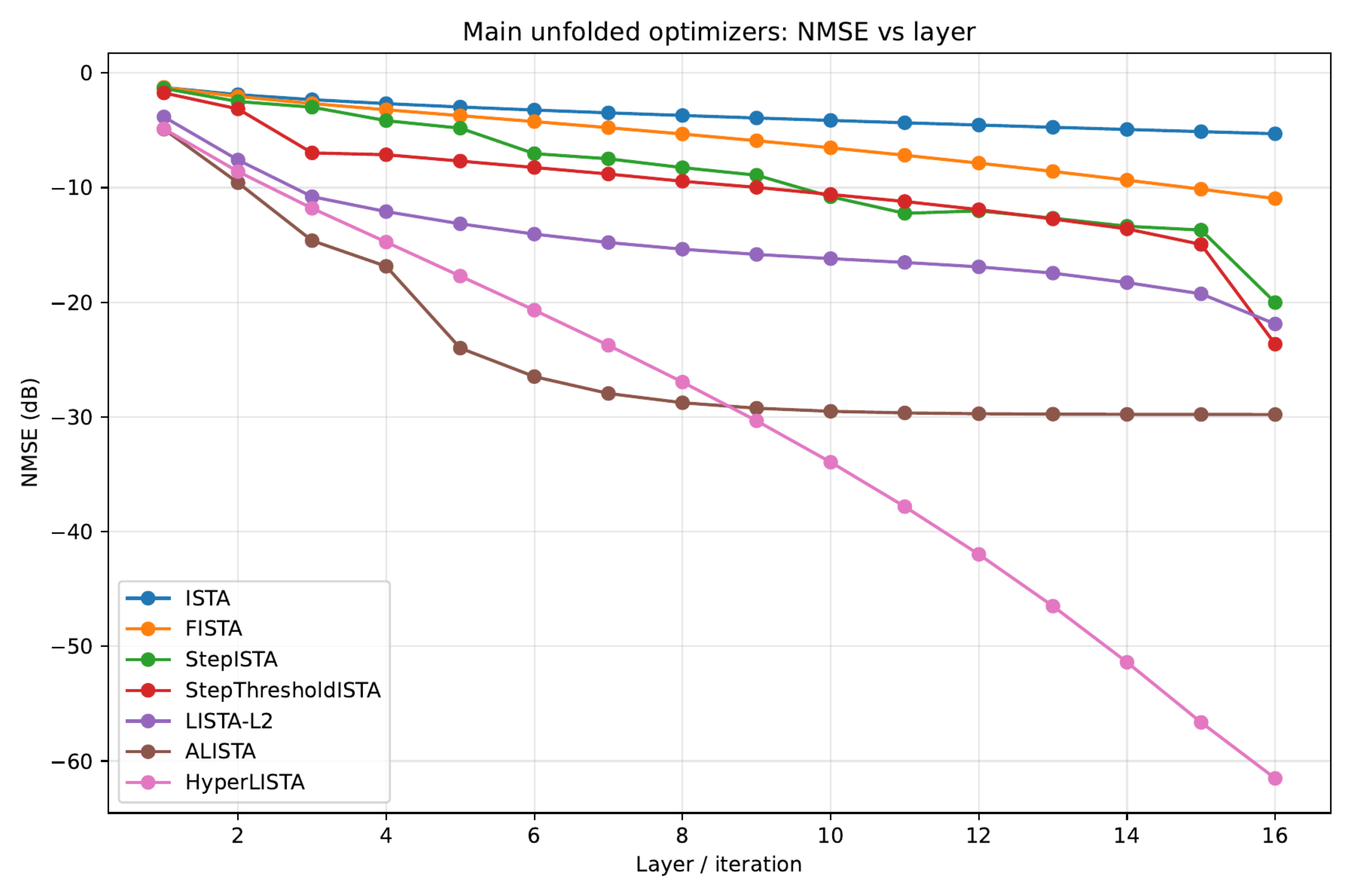
Train layer k while layers 1…k−1 frozen.



Training	NMSE @ K=16	Params
LISTA-L1	−20.3 dB	6 M
LISTA-L2	**−21.9 dB**	**6 M**
LISTA-L3	−16.6 dB	6 M
LISTA-Tied-L1	−19.1 dB	375 K

Tied = one shared (Wx, Wy, θ), RNN-style.
Implicit regularization: −19.1 dB with 16× fewer parameters.

Synthetic Sparse Recovery Results

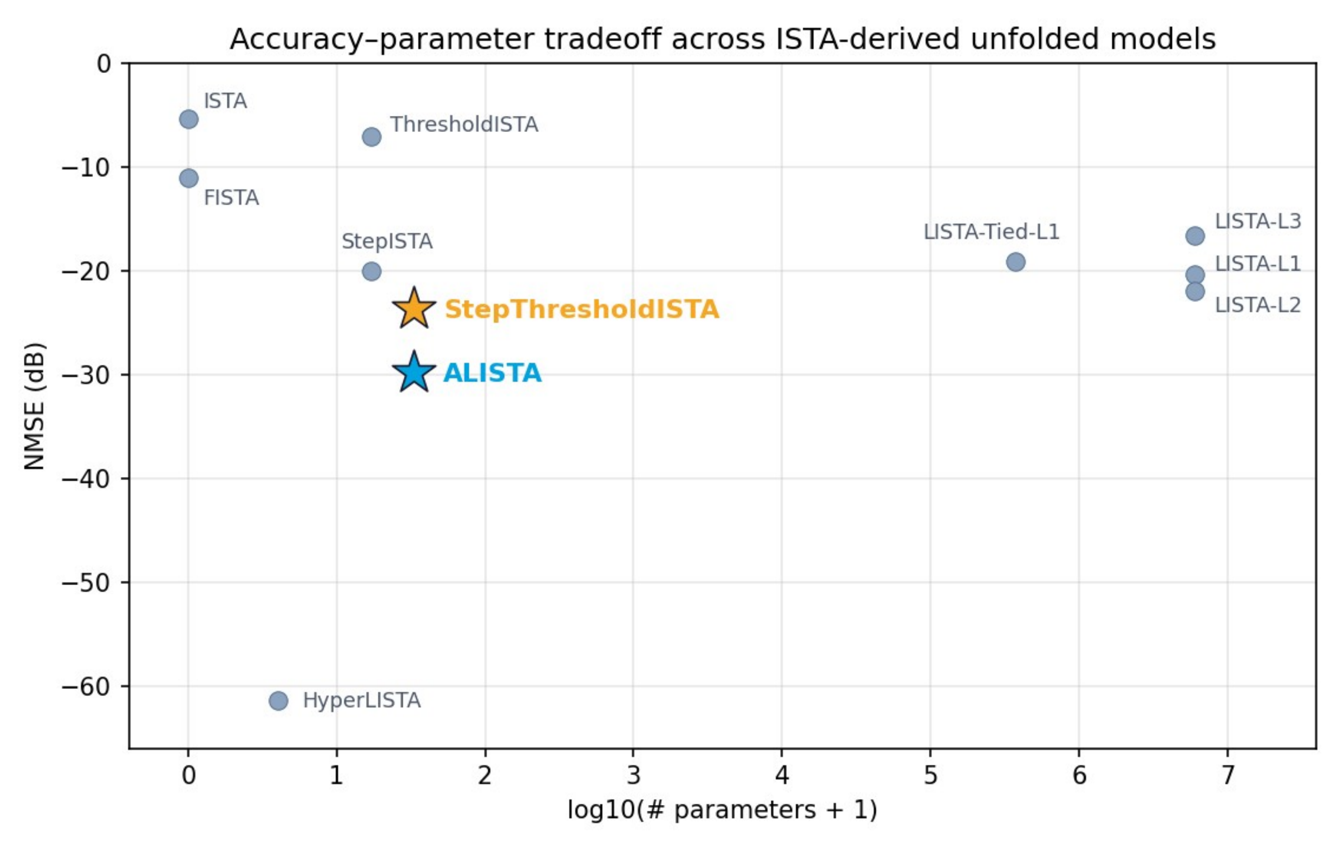


Method	NMSE	Params
ISTA	−5.3 dB	0
FISTA	−11.0 dB	0
StepISTA (ours)	−20.0 dB	16
StepThresholdISTA (ours)	−23.6 dB	32
LISTA-L2 (best LISTA)	−21.9 dB	6 M
ALISTA	**−29.8 dB**	**32**
HyperLISTA	**−61.4 dB**	**3**

Each family is shown with its best variant (LISTA → L2; full L1/L2/L3 comparison on the previous slide).

ALISTA (32 params) beats every LISTA variant; HyperLISTA (3) beats everything.

Design-Space Map



Model	Params	NMSE	What drives it
ThresholdISTA	16	−7.1 dB	learned θ_k
StepISTA	16	−20.0 dB	learned γ_k
StepThresholdISTA	**32**	**−23.6 dB**	$\gamma_k + \theta_k$
LISTA-L2	6 M	−21.9 dB	full matrices + L2
ALISTA	**32**	**−29.8 dB**	**analytic W**
HyperLISTA	**3**	**−61.4 dB**	**adaptive residuals**

StepThresholdISTA (32 scalars)
beats even the best LISTA (L2, 6 M params).
The step-size schedule γ_k is the bottleneck.

Real Image CS: FashionMNIST Pixel Domain

Setup: flatten 28×28 image to $x \in [0,1]^{784}$, measure $b = Ax$, ratio $m/784 \in \{0.25, 0.5\}$.
FashionMNIST: ~51.5% of pixels exactly zero.

Method	NMSE	PSNR	SSIM	Params
ISTA	−1.2 dB	8.9	0.13	0
FISTA	−1.2 dB	8.9	0.14	0
StepThresholdISTA	−1.2 dB	8.9	0.13	32
ALISTA	−0.9 dB	8.4	0.12	32
HyperLISTA	−0.3 dB	7.7	0.09	3
LISTA-L2	**−13.7 dB**	**22.3**	**0.82**	**12 M**

ratio = 0.25

Signal model mismatch

All structured methods assume i.i.d.
Gaussian sparse signals.
FashionMNIST pixels are bounded [0,1],
spatially correlated, and only soft-sparse.

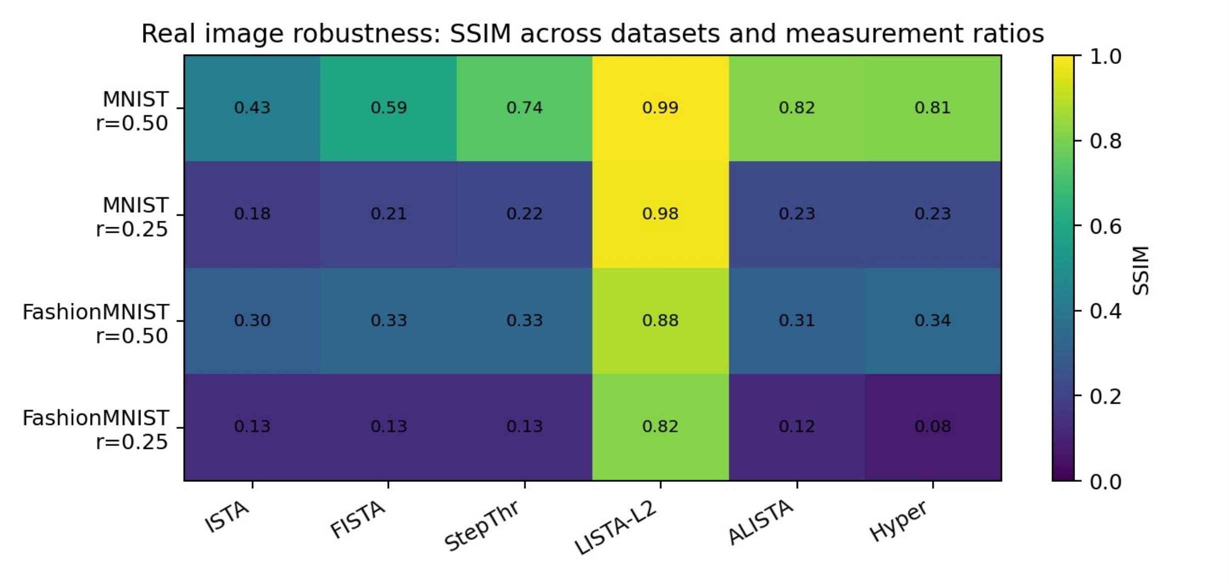
Why LISTA wins

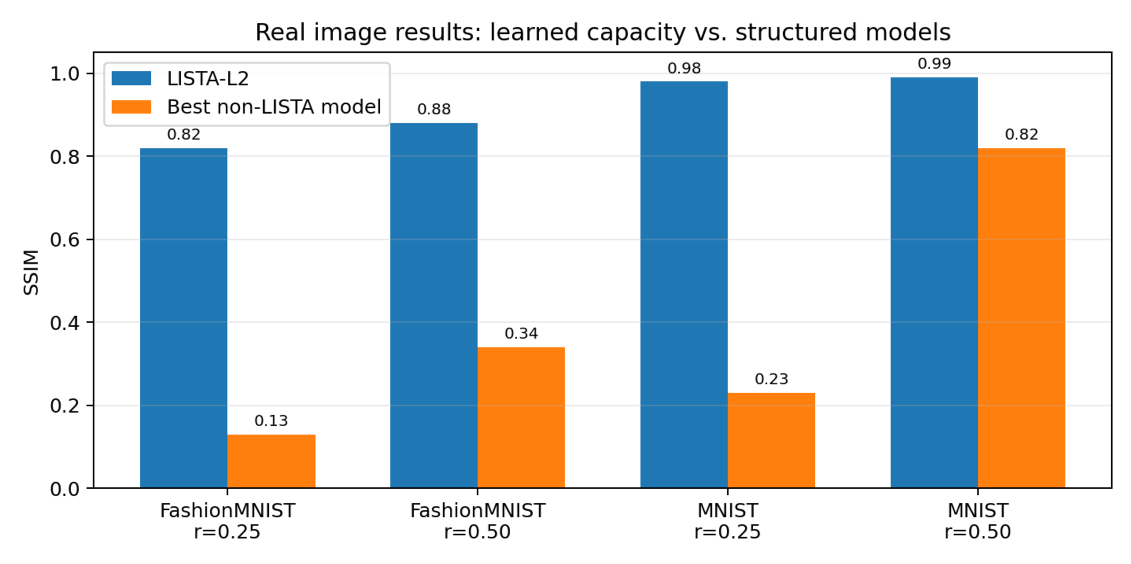
LISTA-L2 is assumption-free —
it learns the actual image distribution
from 51 K training images.

Wrong prior is worse than no prior.

Robustness Across Datasets and Measurement Ratios

SSIM across {MNIST, FashionMNIST} × ratio {0.25, 0.5}: harder data and fewer measurements widen the gap.





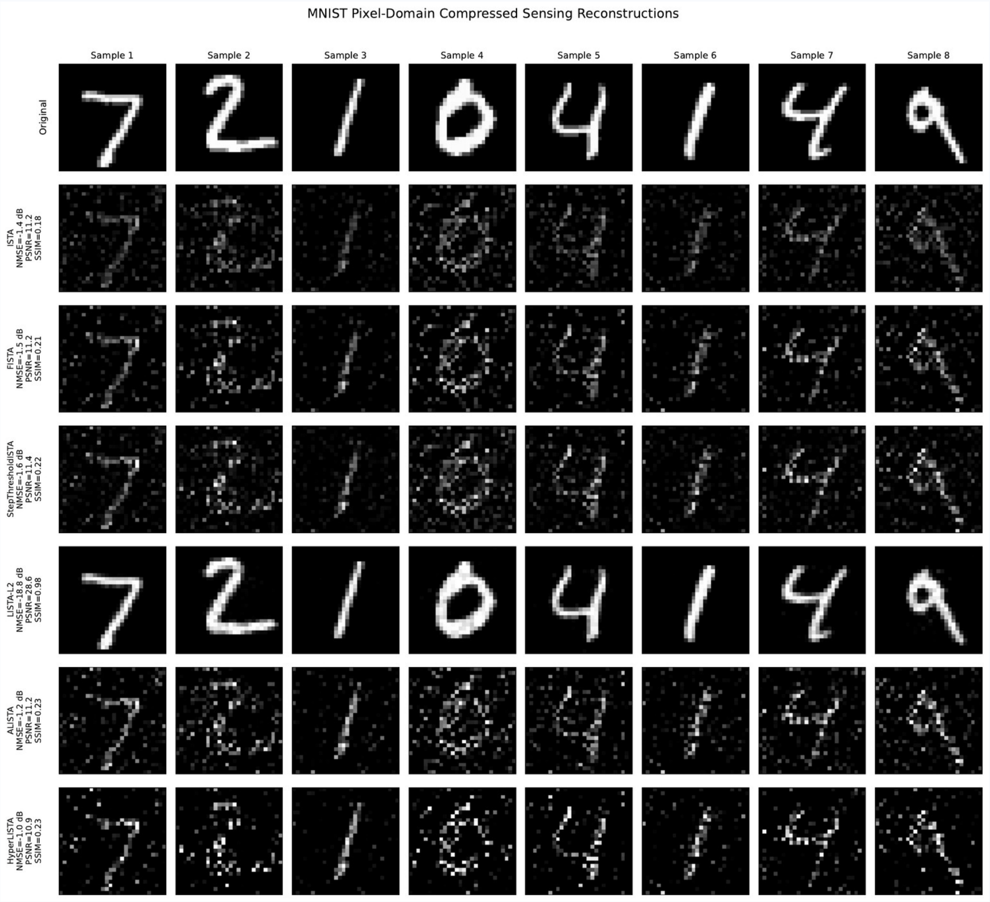
Trend

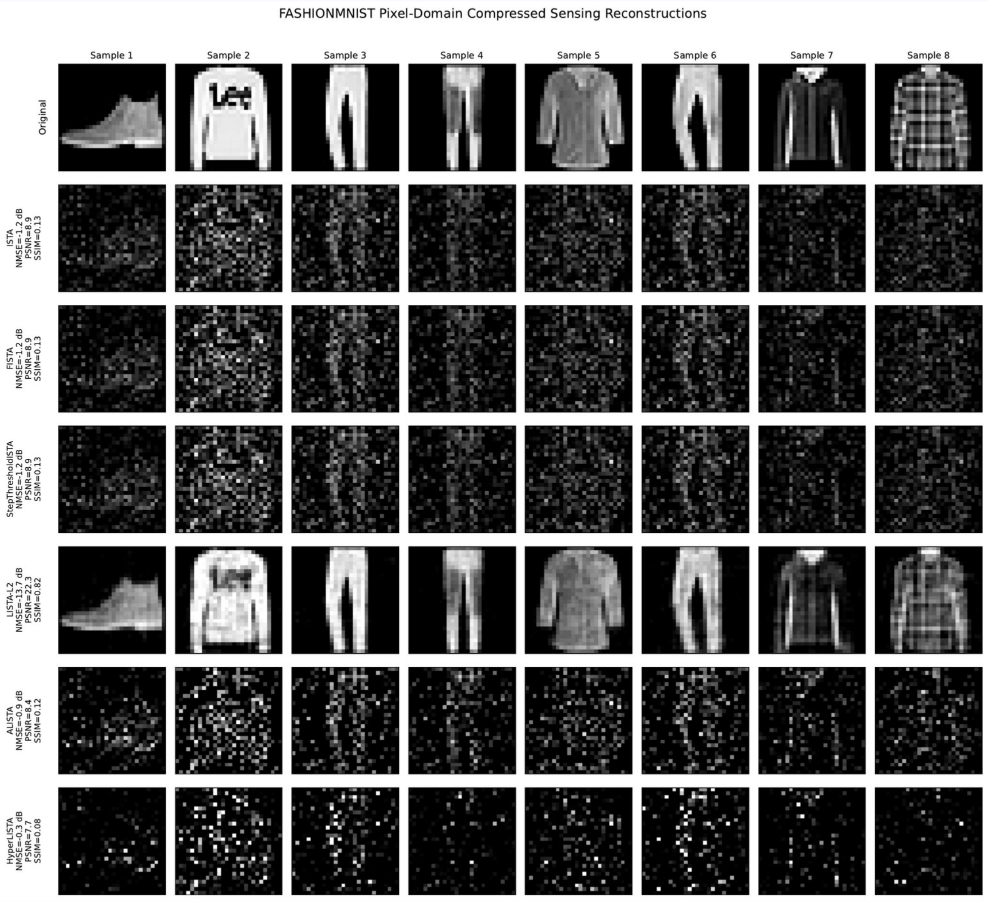
MNIST @ 0.5 is easy for everyone; FashionMNIST @ 0.25 breaks every structured model. LISTA-L2 stays ≥ 0.82 SSIM everywhere.

Interpretation

The sparse-Gaussian prior degrades gracefully on MNIST (~78% zeros) but fails on FashionMNIST (~51% zeros, textured).

Visual Stress Tests (ratio = 0.25)





LISTA-L2 recovers clean structure on both datasets. ALISTA / HyperLISTA zero out valid signal on FashionMNIST — the Gaussian-sparse threshold is miscalibrated for textured images. Wrong prior is worse than no prior.

Conclusions

1 **Structure wins when assumptions match.**

HyperLISTA's 3 scalars beat 6 M learned parameters by 41 dB on synthetic sparse data.

2 **Step sizes matter most.**

StepISTA (16 scalars) matches LISTA. Decoupling θ_k adds 3 dB. Matrix learning is largely redundant.

3 **Weight tying is implicit regularization.**

LISTA-Tied matches LISTA-Untied (−19.1 vs −20.3 dB) with 16× fewer parameters.

4 **Wrong prior actively hurts.**

ALISTA / HyperLISTA score below ISTA on real images. LISTA wins by making no structural assumptions.

MBDL hierarchy on synthetic data:
ISTA → LISTA → ALISTA → HyperLISTA
0 → 6M → 32 → 3 params
−5 → −20 → −30 → −61 dB

Future work

Apply models in DCT / wavelet domain — images are sparser there, closer to the Gaussian-sparse model.

Design image-aware thresholds for bounded, non-negative pixel structure.

Code & References

Code — public GitHub repository (notebooks 01–04 + src modules):

github.com/ThEpiCake/hyperlista-mbdl

References:

[1] A. Beck and M. Teboulle, "A Fast Iterative Shrinkage-Thresholding Algorithm for Linear Inverse Problems," SIAM J. Imaging Sciences, 2009.

[2] K. Gregor and Y. LeCun, "Learning Fast Approximations of Sparse Coding," ICML 2010.

[3] X. Chen, J. Liu, Z. Wang, W. Yin, "Theoretical Linear Convergence of Unfolded ISTA and Its Practical Weights and Thresholds," NeurIPS 2018.

[4] J. Liu, X. Chen, Z. Wang, W. Yin, "ALISTA: Analytic Weights Are As Good As Learned Weights in LISTA," ICLR 2019.

[5] X. Chen, J. Liu, Z. Wang, W. Yin, "Hyperparameter Tuning is All You Need for LISTA," NeurIPS 2021.

[6] N. Shlezinger and Y. C. Eldar, "Model-Based Deep Learning," Foundations and Trends in Signal Processing, vol. 17, no. 4, 2023.

[7] Course lecture notes 1–8, Model-Based Deep Learning (361.2.2320), Ben-Gurion University, 2026.

Thank you very much for listening!